Activity 4.1: Similar Figures & Proportions

A

Name: _____ Partner's name: _____ Date: _____

Directions: Find the value asked for. Write fractions in simplest form. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Solve: $\frac{x}{4} = \frac{9}{6}$

Answer: _____ Match? ☐ Yes ☐ No

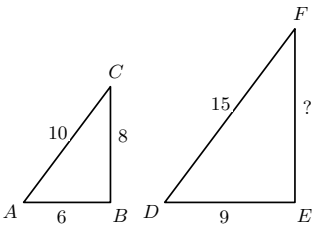
2

A rectangle with sides 4 and 6 is similar to a rectangle with sides 10 and 15. Find the scale factor from the small rectangle to the large one.

Answer: _____ Match? ☐ Yes ☐ No

3

$\triangle ABC \sim \triangle DEF$. Find EF.



Answer: _____ Match? ☐ Yes ☐ No

4

Two similar triangles have a scale factor of 2 : 3. The smaller triangle has perimeter 24. Find the perimeter of the larger triangle.

Answer: _____ Match? ☐ Yes ☐ No

5

Two similar figures have a scale factor of 2 : 3. The smaller figure has area 20. Find the area of the larger figure.

Answer: _____ Match? ☐ Yes ☐ No

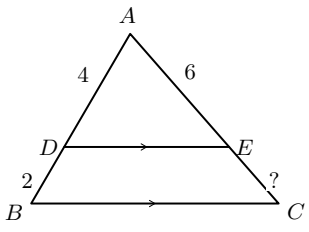
6

A 6-ft person casts a 4-ft shadow. At the same time, a tree casts a 12-ft shadow. How tall is the tree?

Answer: _____ Match? ☐ Yes ☐ No

7

$\overline{DE} \parallel \overline{BC}$. Find EC.



Answer: _____ Match? ☐ Yes ☐ No

8

Two similar polygons have perimeters 40 and 60. A side of the smaller polygon is 12. Find the corresponding side of the larger polygon.

Answer: _____ Match? ☐ Yes ☐ No

Activity 4.1: Similar Figures & Proportions

B

Name: _____ Partner's name: _____ Date: _____

Directions: Find the value asked for. Write fractions in simplest form. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Solve: $\frac{x}{10} = \frac{3}{5}$

Answer: _____ Match? ☐ Yes ☐ No

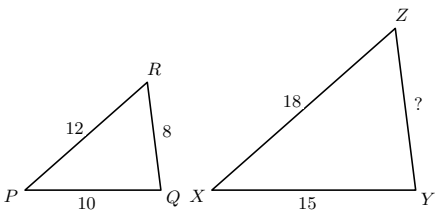
2

A rectangle with sides 6 and 8 is similar to a rectangle with sides 15 and 20. Find the scale factor from the small rectangle to the large one.

Answer: _____ Match? ☐ Yes ☐ No

3

$\triangle PQR \sim \triangle XYZ$. Find YZ.



Answer: _____ Match? ☐ Yes ☐ No

4

The sides of two similar triangles are in the ratio 1 : 3. The smaller triangle has perimeter 12. Find the perimeter of the larger triangle.

Answer: _____ Match? ☐ Yes ☐ No

5

Two similar figures have a scale factor of 1 : 3. The smaller figure has area 5. Find the area of the larger figure.

Answer: _____ Match? ☐ Yes ☐ No

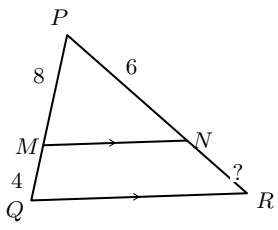
6

A 3-ft stick casts a 2-ft shadow. At the same time, a flagpole casts a 12-ft shadow. How tall is the flagpole?

Answer: _____ Match? ☐ Yes ☐ No

7

$\overline{MN} \parallel \overline{QR}$. Find NR.



Answer: _____ Match? ☐ Yes ☐ No

8

A photo that is 8 in. by 12 in. is enlarged so that its short side is 12 in. Find the long side of the enlargement.

Answer: _____ Match? ☐ Yes ☐ No